

Regulation of mechanical properties of diene rubber cured by oxa-Michael Reaction via manipulating network structure

Xuhui Zhang, Zhenghai Tang, Baochun Guo*

Department of Polymer Materials and Engineering, South China University of Technology, Guangzhou, 510640, China

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ABSTRACT

As diene rubber products are generally fabricated with complicated curing package involving toxic additives, the exploration of alternative green curing chemistry for diene rubber is of great importance. In this contribution, we demonstrate a curing chemistry based on oxa-Michael reaction for diene rubber to achieve the combination of simple curing system and tunable mechanical properties. Specifically, pendent hydroxyls are introduced onto the solution-polymerized styrene-butadiene rubber (SSBR) via thiol-ene click reaction and the modified SSBR is effectively cured by various acrylates without the use of any additional additives. The mechanical behaviors of vulcanizates can be widely regulated by manipulating the network structures, which is tuned by curing temperature, hydroxyl content of SSBR, content and functionality of acrylates. The correlation between mechanical properties and network structure is accordingly achieved, which provides essential basis for the future application of the diene rubbers cured by oxa-Michael reaction.

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1. Introduction

Diene rubbers are widely used in diverse industrial applications such as tires, seals and dampers due to their high extensibility and elasticity [1–3]. Curing of the rubbers into three-dimensional network is essential for achieving high elasticity. In most cases, diene rubbers are cured by sulfur-based curing packages or peroxides [4,5]. However, sulfur-cured rubbers suffer from some significant deficiencies. Generally, different activators and accelerators are essential for efficient crosslinking due to that when activators and accelerators are absent, sulfur exhibits a rather slow crosslinking rate and sulfur couples onto rubber chains mainly in the form of side cyclic structures [6,7]. The common activators are zinc oxide and stearic acid while the accelerators can be dibenzothiazolodisulfide (DM), N-cyclohexyl-2-benzothiazole sulphonamide (CZ), diphenylguanidine (D) or a combination of them [8]. In most cases, anti-ageing reagents, such as N-isopropyl-N'-phenyl-4-phenylenediamine (IPPD) and N-1,3-dimethylbutyl-N'-phenyl-4-phenylenediamine (DMPPD), are also required. Those necessary assistants make the sulfur-based curing package complicated. Notably, many of the assistants are toxic [9–11]. For example,

according to the dangerous substances directive of European Union laws (67/548/EEC), accelerant CZ and accelerant DM are classified as R43 (may cause sensitization by skin contact) and R50/53 (very toxic to aquatic organisms) [9]. Amine-based anti-ageing agents, IPPD and DMPPD, may cause contact dermatitis and be harmful to the environment [10,11]. Therefore, exploration of effective yet simple curing system for diene rubbers is of great importance.

Oxa-Michael reaction, belonging to hetero-Michael reactions, is the addition reactions between oxygen nucleophiles and conjugated systems [12]. Due to the wide applications in organic syntheses of valuable intermediates, especially in the syntheses of oxygen-containing heterocycles, oxa-Michael reaction has drawn great attentions in recent years [13–17]. The oxa-Michael reaction has also been successfully applied in the polymerization of hydroethyl acrylate [18,19] and the preparation of hyperbranched polymer via proton-transfer polymerization of acrylate monomers with two hydroxyls [20]. Very recently, our lab revealed that epoxidized natural rubber (ENR) could be effectively cured via oxa-Michael reaction, by using unsaturated carboxylates as the crosslinkers. The ENR vulcanizates exhibited good mechanical properties as well as excellent ageing resistance [9,21]. We have also demonstrated the shape memory effects of ENR cured via oxa-Michael reaction [22–24]. Therefore, this new crosslinking chemistry broadened the gallery of rubber materials. Still, there are some unsolved issues for crosslinking via oxa-Michael reactions. The systems are limited to

* Corresponding author.

E-mail address: psbcguo@scut.edu.cn (B. Guo).

ENR-based elastomers, and insufficient curing activity as evidenced by long-marching curing behaviors has been experienced [9,21,25]. More importantly, how the network structure affects the mechanical properties remains unclear. Consequently, exploring new rubber systems and understanding the relationship between the network structures and mechanical properties are significant for future applications.

In this contribution, we demonstrate the crosslinking via oxo-Michael reaction between hydroxylated-diene rubbers and acrylates. The starting diene rubber, a solution-polymerized styrene-butadiene rubber (SSBR), is firstly functionalized via thiol-ene chemistry with mercaptoethanol (ME) to produce SSBR with pendent hydroxyls (OSx). With no additional additives, OSx can be effectively crosslinked via acrylates. The network structures are manipulated by hydroxyl content, crosslinker content, functionality of crosslinkers and curing conditions. Consequently, the mechanical properties can be regulated via manipulating the network structure and hence the correlation between them is achieved.

2. Experimental section

2.1. Materials

The Solution-polymerized styrene-butadiene rubber (SSBR 2506, styrene content of 20.2 wt.%, vinyl content of 44.6 wt.%, $M_n = 3.95 \times 10^5$, and $M_w/M_n = 1.15$) was kindly provided by Yanshan Petrochemical Co. Sinopec. 2-Mercaptoethanol (ME) and diethylene glycol diacrylate (A^2) were purchased from Alfa Aesar (China) Chemicals Co., Ltd. Trimethylolpropane triacrylate (A^3) and pentaerythritol tetraacrylate (A^4) were purchased from Aladdin (Shanghai) and TCI (Shanghai) Co., Ltd., respectively. Azobisisobutyronitrile (AIBN) was purchased from Sigma-Aldrich. AIBN was recrystallized in 95% ethanol (1 g/10 mL) and stored at 4 °C. Other rubber additives were industrially available products and used as received without further purification. The structures of A^2 , A^3 , A^4 are depicted in Scheme 1.

2.2. Preparation of hydroxylated SSBR (OSx)

In a three-necked flask, 70 g SSBR was well dissolved in toluene (0.067 g/mL). The temperature of the rubber solution was elevated to 80 °C under nitrogen purging and stirring. Pre-determined amounts of ME and AIBN (0.19 g, 0.2 mol% of vinyl) were dissolved in 20 mL toluene, respectively. ME and AIBN solutions are dropped into the SSBR solution successively. The reaction was lasted for 2 h at 80 °C. The resultant solution was poured into methanol solution under stirring to precipitate the modified SSBR. The product was further dissolved in toluene and precipitated by methanol repeatedly to fully remove the unreacted agents.

Subsequently, the resulting polymer was dried in a vacuum oven at 50 °C for 2 days to remove the residual solvent. The hydroxylated SSBR is denominated as OSx, where x represents the mole ratio between grafted ME and vinyl.

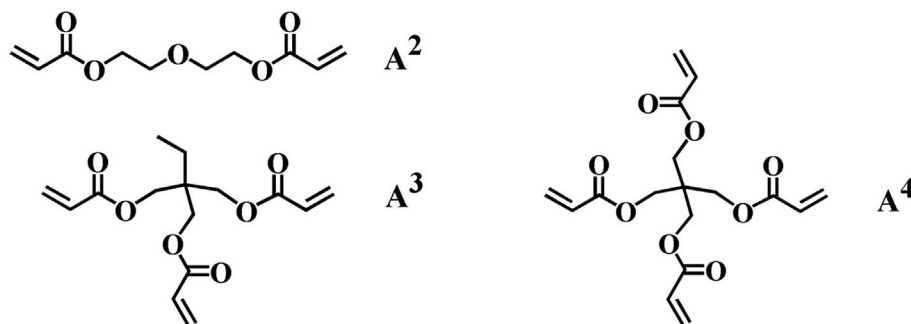
2.3. Preparation of cured samples

A desired amount of OSx were dissolved into toluene (0.067 g/mL). Subsequently, pre-determined amounts of crosslinks (toluene solution, 0.02 g/mL), was added into OSx solution. The mixture is stirred for 5 h and then dried. The dried mixtures were subjected to compression moulding at 160 °C for the optimum curing time, which are determined by a vulcameter. The cured samples are named as OSx A^w_z , where w and z are the functionality and content (phr, parts per hundreds of rubber) of the crosslinker, respectively. Specially, sulfur-cured OS7.8 was also prepared as the control sample. The basic recipe for the control sample is as follows: SSBR 100 g; zinc oxide 5 g; stearic acid 1 g; dibenzothiazolodisulfide (DM) 0.5 g; N-cyclohexyl-2-benzothiazole sulphonamide (CZ) 1.5 g; and sulfur 1.5 g. The control sample was also subjected to compression moulding at 160 °C for the optimum curing time.

2.4. Characterization

Proton nuclear magnetic resonance (^1H NMR) spectra were recorded on a Bruker AVANCE III 400 MHz spectrometer. The samples were dissolved in CDCl_3 and TMS was used as an internal standard. Fourier transform Infrared spectroscopy (FTIR) in attenuated total reflectance (ATR) mode was performed on a Bruker Vertex spectrometer. The scan range is from 3600 to 600 cm^{-1} with a resolution of 4 cm^{-1} . Differential scanning calorimetry (DSC) was recorded on a TA Q20 instrument (USA). The samples were heated to 80 °C to eliminate any thermal histories. The samples were then cooled to -80 °C, and reheated at 10 °C/min to measure the glass transition temperature (T_g) of OSx. Curing profiles of all compounds were recorded at pre-determined temperature by a U-CAN UR-2030 vulcameter (Taiwan) according to ASTM D 5289. All the curing parameters are summarized in Table S1. Tensile tests at room temperature were measured using a UCAN UT-2060 (Taiwan) instrument following ASTM D 412. The tensile sample is a dumbbell shape with a typical dimension of 115 mm $\times$ 6 mm $\times$ 0.6 mm manufactured by corresponding cutting machine. The extension rate is 500 mm/min. Dynamic mechanical analysis (DMA) in tensile mode was performed on a TA Q800 machine. The scanning temperature is ranged from -80 to 100 °C. The frequency and heating rate were set as 1 Hz and 3 °C/min, respectively.

Crosslink density was measured by the equilibrium swelling method [26]. Vulcanizates were firstly weighted and swollen in toluene at room temperature for 72 h. Subsequently, the sample is



Scheme 1. Molecular structures of crosslinkers with different functionality.

removed from the solvent and weighed after wiping off the surface toluene. The samples were further dried in a vacuum oven for 36 h at 80 °C to remove the solvent. After drying, the sample is reweighed. The volume fraction (V_r) of OSx in the swollen gel was calculated by Eq. (1) [27].

$$V_r = \frac{m_0 \times \varnothing \times (1 - \alpha) / \rho_r}{m_0 \times \varnothing \times (1 - \alpha) / \rho_r + (m_1 - m_2) / \rho_s} \quad (1)$$

where m_0 is the sample mass before swelling; m_1 and m_2 are sample masses before and after drying respectively; $\varnothing$ is the mass fraction of rubber in the vulcanizates; α is the mass loss of the vulcanizate during swelling; and ρ_r and ρ_s are the rubber and solvent density, respectively. The crosslink density, V_e , was then calculated by Eq. (2) (Flory-Rehner equation) [28].

$$V_e = -\frac{\ln(1 - V_r) + V_r + \chi V_r^2}{V_s(V_r^{1/3} - 2V_r/f)} \quad (2)$$

where V_r is the volume fraction of OSx in the swollen gel and V_s is the solvent molar volume (106.5 cm³/mol for toluene). χ is the SSBR-toluene interaction parameter and is taken as 0.0653 according to reference. [26] f is the functionality of interlinkage, which is four for sulfur cured non-end crosslinked diene rubbers while it is 4, 6 and 8 for A², A³, A⁴ respectively.

3. Results and discussion

3.1. Functionalization of SSBR via thiol-ene reaction

A series of OSx were prepared based on the thiol-ene addition reaction between vinyls on SSBR and mercaptan (-SH) on ME (Scheme 2) [29]. The effectiveness of the grafting process is initially verified by FTIR spectra (Fig. 1). The intensity of the peak around 699 cm⁻¹, characterizing the bending vibration of monosubstituted benzene [30], is normalized to compare the changes of other moieties. The peaks around 909 cm⁻¹ and 994 cm⁻¹ correspond to the out-of-plane bending vibration of =CH₂. Clearly, with increasing ME dose, the intensities of the two peaks continuously decrease, indicating the consumption of pendent vinyl group in SSBR.

¹H NMR was also performed to demonstrate the thiol-ene addition reaction (Fig. 2). Peaks centered at 4.9 ppm and 5.3 ppm are related to the signal of protons in CH₂= and =CH- of vinyl, respectively. Compared with the spectrum of neat SSBR, OS 10.1 exhibits a decrease of both signals at 4.9 ppm and 5.3 ppm, demonstrating the conversion of vinyl groups. Besides, new signal appears around 2.6 ppm in the modified SSBR, which is assigned to the protons in -CH₂-S-CH₂-. In addition, another newly generated signal around 3.6 ppm is attributed to the protons of -CH₂-O-. The emerging of these two new signals indicates the successful grafting

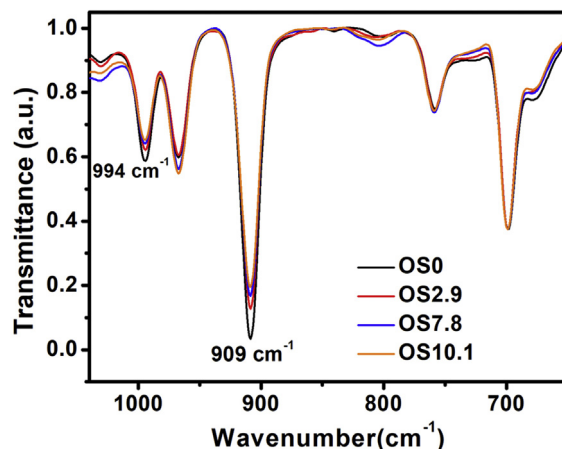


Fig. 1. Evolution of FTIR spectra of OSx with increasing ME dose.

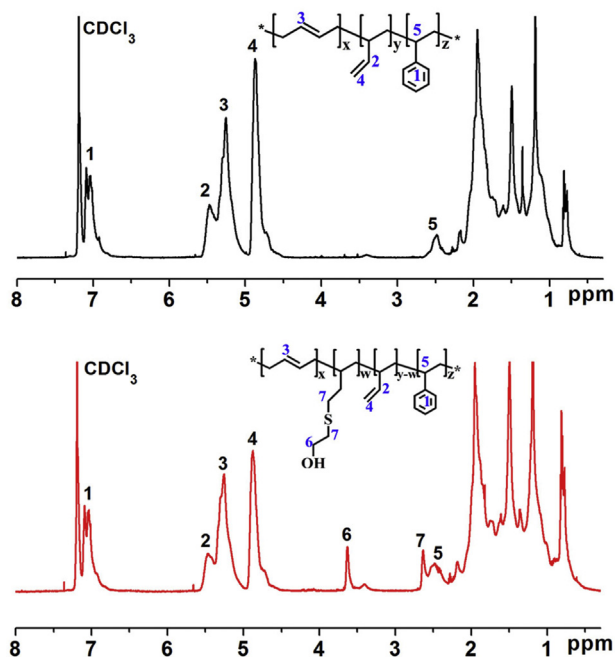
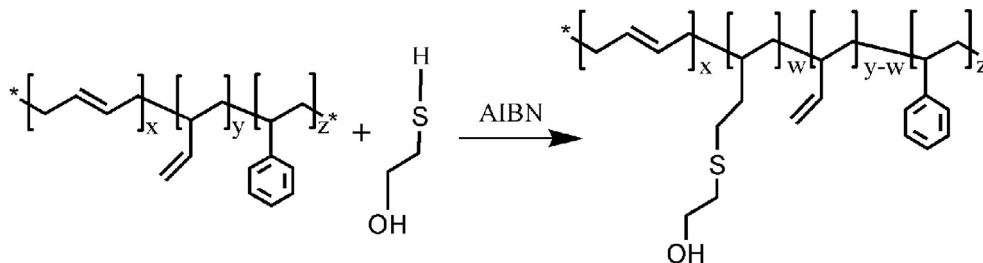


Fig. 2. ¹H NMR spectra of neat SSBR and OS10.1.

of -OH onto SSBR. The ¹H NMR spectra of OS2.9 and OS7.8 exhibit similar but weaker signal changes than OS10.1 (Figs. S1 and S2). All those evidences are consistent with previous reports [31–33] and convincingly demonstrate the successful graft of ME onto SSBR. In addition, the integral areas of protons before and after reaction



Scheme 2. Thiol-ene addition between ME and SSBR.

allow us to calculate the molar ratio between grafted ME and vinyl (R_g) from Eq. (3), and the detailed parameters of OSx are listed in Table 1.

$$R_g = \frac{I_{2.6, x}/4}{I_{4.9, 0}/2} \times 100\% = \frac{I_{2.6, x}}{2I_{4.9, 0}} \times 100\% \quad (3)$$

where $I_{2.6, x}$ is the integral area of new signal around 2.6 ppm for OSx; $I_{4.9, 0}$ is the integration of signal centered at 4.9 ppm of neat SSBR.

DSC of OSx were conducted to investigate the effect of R_g on the glass transition temperatures (T_g s) of the gums. As shown in Fig. 3, only a single glass transition is observed, indicating the amorphous nature of the modified SSBR. With the increment of R_g , T_g increases continuously from -26.1 to -17.7 °C. This phenomenon should be ascribed to the hydrogen bonding between the pendent -OH, which suppresses the chain motion, resulting in elevated T_g s. Besides, all the measured T_g s are much lower than room temperature, indicating good cold tolerance.

3.2. Crosslinking between OSx and Acrylates based on Oxa-Michael Reaction

To verify the occurrence of oxa-Michael reaction between OS10.1 and acrylates, we monitored the FTIR spectra of A^3 , OS10.1, OS10.1/ A^3 compound and OS10.1/ A^3 vulcanizate (Fig. 4). For A^3 , the peaks around 1409, 1050 and 810 cm^{-1} are ascribed to the deformation vibration, rocking vibration and twisting vibration of $=\text{CH}_2$, respectively, while the peak at 1640 cm^{-1} is ascribed to the stretching vibration of $\text{C}=\text{C}$ [34,35]. The peak at 1178 cm^{-1} is related to the stretching vibration of $\text{C}-\text{O}$ in esters [34,35]. For OS10.1/ A^3 compound, the peak around 1178 cm^{-1} is narrow and intensive. After curing, this peak is widened and the $\text{C}-\text{O}$ signal exhibits a blue shift to 1156 cm^{-1} , which is assigned to $\text{C}-\text{O}$ stretching vibration in aliphatic ether. The widening of the peak should be ascribed to the merging of the two peaks belonging to $\text{C}-\text{O}$ stretching vibrations. It is worth noting that compared with OS10.1/ A^3 compound, the intensity of characteristic peaks of A^3 in OS10.1/ A^3 vulcanizate exhibits a sharp decrease. There are two reasons contributing to this phenomenon. On one hand, the increased compatibility between A^3 and OS10.1 rising from the crosslinking reaction should be mainly responsible for the sharply decreased intensity. Due to the occurrence of oxa-Michael reaction, part of A^3 is chemically involved into the rubber matrix, resulting in partial shielding of signals belonging to A^3 . On the other hand, $\text{C}=\text{C}$ in A^3 is reactive and will be consumed during the crosslinking. All the evidences imply the oxa-Michael reaction between hydroxyl on OS10.1 and acryloyloxy on A^3 .

The curing behaviors of OSx/ A^3 compounds were also examined to verify the effectiveness of the crosslinking mechanism. Fig. 5 shows the curing profiles of A^3 -cured OS7.8 under different temperature. One can clearly find that A^3 -cured OS7.8 exhibit a classic curing behavior, indicating the high curing efficiency of A^3 to OS7.8. With the increasing of curing temperature, the curing rate of OS7.8/ A^3 increases consistently, resulting in a shorter curing time. When temperature reaches 160 °C, the curing profile can reach an

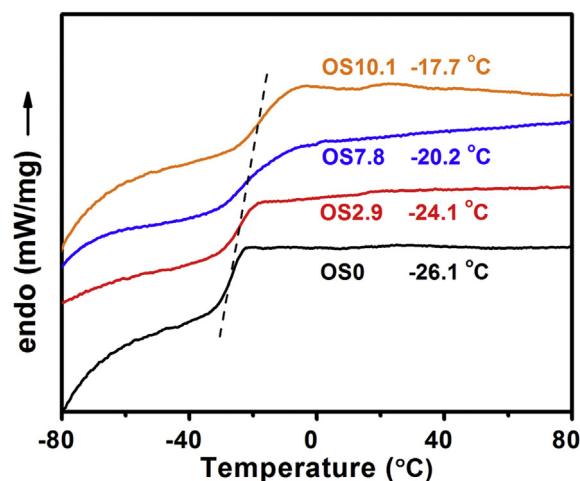


Fig. 3. DSC curves of OSx for the second heating run.

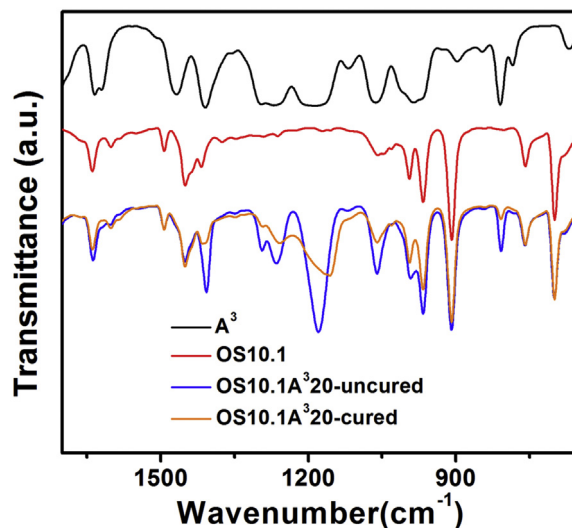


Fig. 4. FTIR spectra of A^3 , OS10.1, OS10.1/ A^3 compound and OS10.1/ A^3 vulcanizate.

equilibrium within 20 min, which is comparable with traditional sulfur-based curing systems. Besides, the maximum torques (M_H) are almost the same under different curing temperature, indicating that complete crosslinking can be achieved at different temperatures.

Based on the above discussion, we demonstrate the occurrence of crosslinking chemistry based on oxa-Michael reaction (Scheme 3). Although oxa-Michael reactions between alcohols (small molecule) and α , β -unsaturated ketones have been widely investigated, this reaction has not been applied in crosslinking hydroxyl-containing diene rubber. Herein, under elevated temperature, -OH is activated and attacks the $\text{C}=\text{C}$ in acrylates, followed by proton-transfer to complete the oxa-Michael reaction. Generally, sulfur-

Table 1
Summary of ingredients ratio and graft ratio of the reaction.

Sample	Molar ratio ME/vinyl/AIBN	R_g^a	Hydroxyl concentration (mmol/100 g gum)
OS2.9	0.03/1/0.002	2.9	24
OS7.8	0.09/1/0.002	7.8	64
OS10.1	0.15/1/0.002	10.1	83

^a R_g refers to the molar ratio between grafted ME and vinyl.

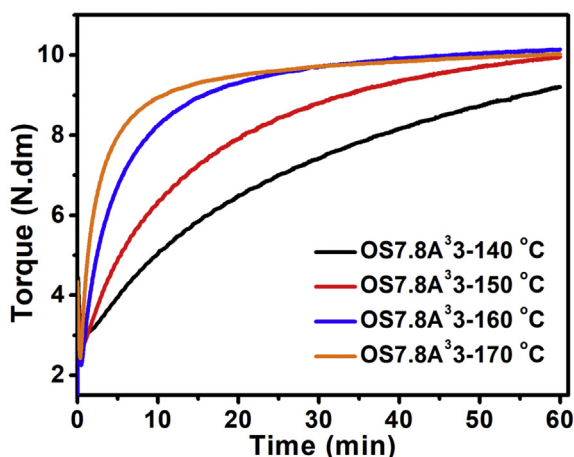


Fig. 5. Curing profiles of A³-cured OSx under different curing temperature.

based crosslinking requires a number of toxic additives, including activators and accelerators. In addition, the amount of these additive can be as high as 8–10 parts related to 100 parts of gum and the toxic additives are usually no less than 1 part per 100 parts of gum [36]. Explicitly, the crosslinking via oxa-Michael reaction only need acrylates, rendering it a clear, simple and less-toxic curing technology when compared with traditional sulfur-based curing package.

3.3. Manipulation of Mechanical Behaviors via Regulating the Network Structure

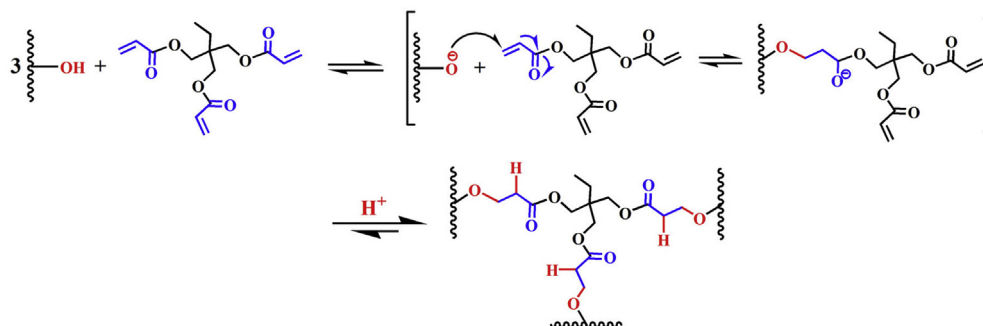
We studied the effect of A³ content on the properties of the vulcanizates. Fig. 6a is the histogram of crosslink density of A³-cured OS7.8. Apparently, the crosslink density increases monomously with increasing A³ dose due to the increment of reactive points. For example, the crosslink density increased from 2.2×10^{-4} mol/cm³ when the dose of A³ is increased from 1.5 phr to 5 phr. Noticeably, the crosslink density of OS7.8A³3 (2.9×10^{-4} mol/cm³) is comparable to that of sulfur-cured OS7.8 (Fig. 6a). Fig. 6b depicts the curing curves of OS7.8 cured by different amount of A³. With the increment of A³ content, M_H increases continuously due to the increased crosslink density. Besides, the crosslinker contents have little effect on curing time. The stress-strain curves of A³-cured OSx and sulfur-cured OS7.8 are shown in Fig. 6c. With the increasing of A³ dose, the tensile strength and modulus of A³-cured OS7.8 increase obviously while the elongation at break is compromised, which should be ascribed to the increased crosslink density. Specifically, the modulus at 200% strain of OS7.8A³5 is as high as 4.8 MPa, which is even higher than

that of sulfur-cured SSBR composites containing 50 phr silica [37]. The influence of A³ content on the dynamic behavior of vulcanizates is revealed in Fig. 6d and Fig. S3. One can find that the peak value of tan δ is decreased with increasing A³ content due to the enhanced constraint in chain motion rising from increased crosslink density.

The influence of hydroxyl content of OSx on the properties of vulcanizates are also studied. As shown in Fig. 7a, when the acryloyloxy concentration is constant (30 mmol/100 g gum), the crosslink density increases with increasing hydroxyl content of OSx, which is reasonable as the OSx with higher hydroxyl content offers more curing sites. Although the functionality of A³ is three, due to the high viscosity of the rubber, it is possible that only one acryloyloxy of A³ reacts with rubber chains and the other acryloyloxy groups cannot be accessed by the hydroxyls due to the relatively low hydroxyl content of OS2.9 (24 mmol/100 g gum). Consequently, part of A³ react with the rubber, but the valid crosslink is not formed. At higher hydroxyl content, the acryloyloxy groups of A³ are more accessible to the hydroxyls of rubber chains to form more valid crosslinks. Fig. 7b depicts the curing profiles of A³3-cured OSx. M_H of the samples increases continuously with increasing -OH content due to the improving of crosslink density. The stress-strain curves of A³3-cured OSx are depicted in Fig. 7c. It is clear that the tensile strength and modulus are enhanced with the increment of hydroxyl content. However, the elongation at break is essentially independent of hydroxyl content. Although increasing crosslink density may lead to the improvement in modulus and strength, in this system, the enhanced modulus and strength should be attributed to the improved cohesive energy density rising from the higher hydroxyl content. The influence of -OH content on the dynamic behavior of vulcanizates is revealed in Fig. 7d and Fig. S4. In Fig. 7d, with increasing hydroxyl content, the peak value of tan δ decreases consistently due to improved constraints in segmental mobility. Besides, the T_gs also show an increasing trend with increasing hydroxyl content, which should be mainly ascribed to the different T_g of gum as shown in Fig. 3.

We further investigated the effects of functionality of acrylate on the performance of vulcanizates. In conventionally crosslinked diene rubbers, it is not possible to experimentally tune the network structures via manipulating *f*. Based on oxa-Michael crosslinking mechanism, the *f* can be easily controlled via applying acrylates with different functionality. The modified SSBR, OS7.8, is utilized as the matrix while acrylates with functionality of 2, 3, 4 are applied as crosslinkers. Equal molar quantity of acryloyloxy (3.2 phr A², 3 phr A³ and 2.6 phr A⁴) are added into the matrix to ensure equal reactive points.

Fig. 8a exhibits the increasing trend of crosslink density with increasing *f*. This phenomenon may be ascribed to two main reasons. On one hand, crosslinker with higher functionality is able to arrest more rubber chains at each crosslink in the rubber network,



Scheme 3. Proposed vulcanization mechanism of OSx cured by oxa-Michael reaction.

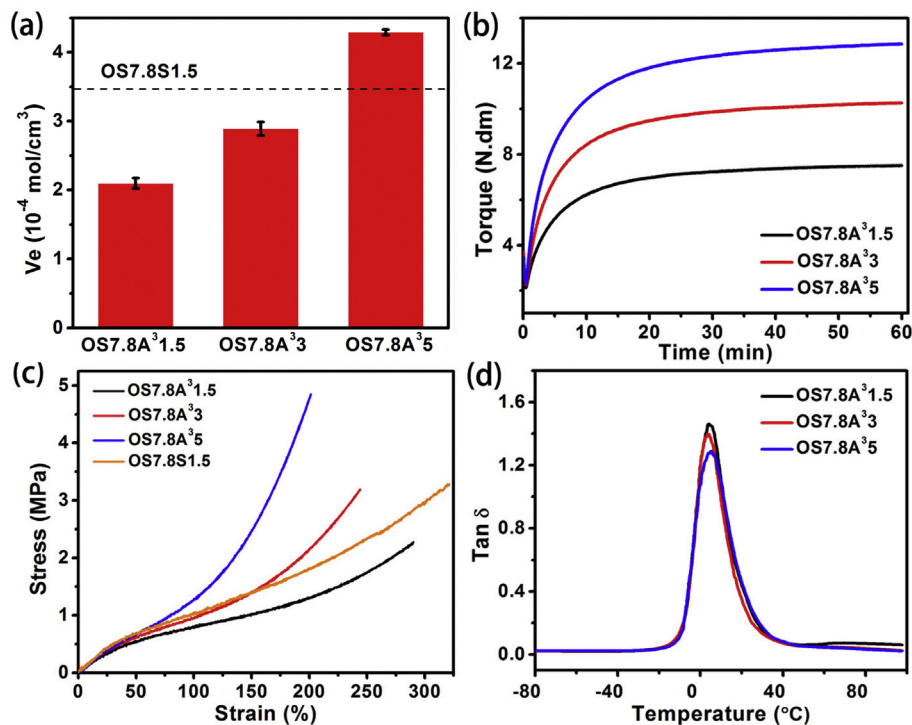


Fig. 6. Crosslink density (a), curing curves (b), stress-strain curves (c) and tan δ -T curves (d) of OS7.8 cured by different amount of A^3 . The dash line in Figure (a) shows the crosslink density of sulfur-cured OS7.8.

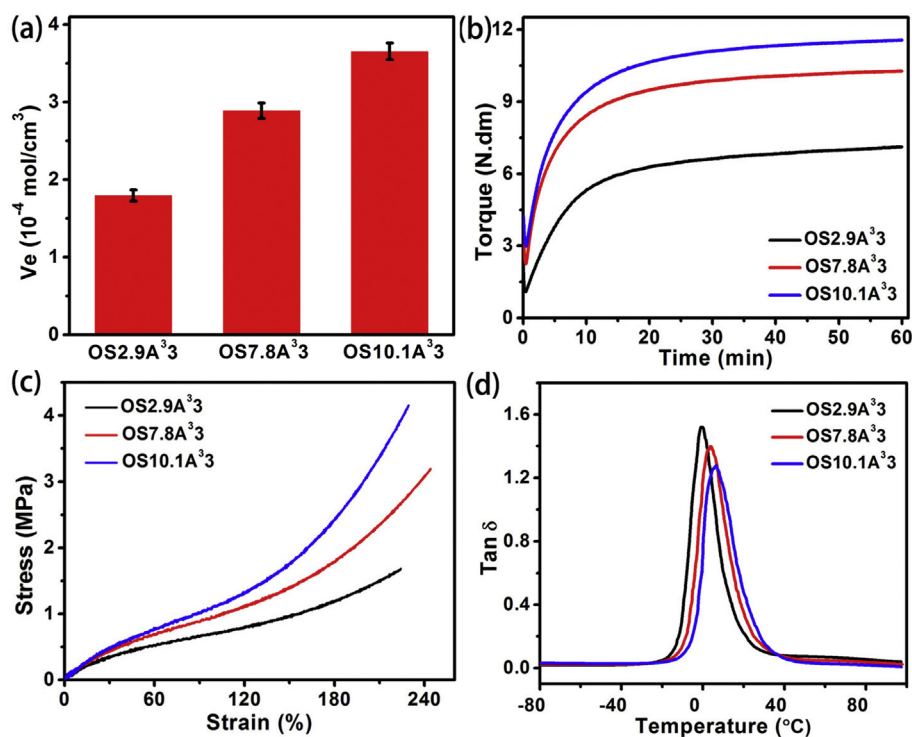


Fig. 7. Crosslink density (a), curing profiles (b), stress-strain curves (c) and tan δ -T curves (d) of OSx cured by 3 phr A^3 .

leading to higher crosslink density. On the other hand, crosslinkers with higher functionality can be more efficiently converted into valid crosslinks. For A^2 , to achieve valid crosslink, two acryloyloxy in A^2 are both required to react with different rubber chains;

otherwise pendent chains or loop will be formed. As for A^4 , any two acryloyloxy groups react with different rubber chains can achieve a valid crosslink. Therefore, rubbers cured by crosslinkers with higher functionality possess higher crosslink density. Fig. 8b shows

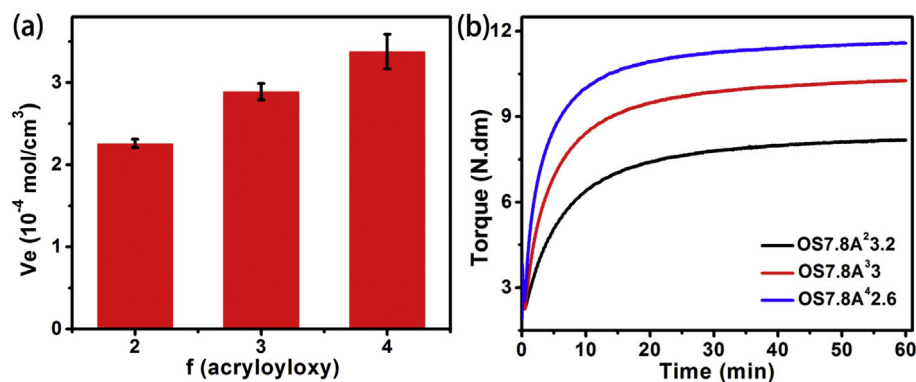


Fig. 8. Crosslink density histograms (a) and curing profiles (b) of OS7.8 cured by acrylates with different functionality.

the curing profiles of OS7.8 cured by A², A³, A⁴. Apparently, all samples exhibit similar and quick curing behaviors, indicating that A² and A⁴ can also effectively cure hydroxylated SSBRs. Conceivably, based on oxa-Michael reaction, acrylates with tunable functionality can crosslink hydroxyl-containing rubbers efficiently. Besides, with the same quantity of reactive points, crosslinkers with higher functionality can provide vulcanizates with higher M_H . For example, the M_H of cured OS7.8 increases obviously from 8.2 N dm to 11.6 N dm as f is altered from 4 to 8. This increase in M_H should be mainly ascribed to the improved crosslink density. Besides, improved crosslink density also results in the slightly decreased peak value in $\tan \delta$ -T curves (Fig. S5).

The typical stress-strain curves of acrylates-cured OS7.8 are shown in Fig. 9a. One can see that with the increment of functionality, the tensile strength and modulus increased monotonously. Strikingly, compared with OS7.8A^{2.3.2}, the tensile modulus (stress at strain of 200%) and tensile strength of OS7.8A^{4.2.6} are increased by 90% and 51%, respectively.

To understand the effect of functionality on segmental behaviors during uniaxial extension, the elastomeric network is estimated by using the Mooney-Rivlin equation (4) [38]:

$$\sigma^* = \frac{\sigma}{\lambda - \lambda^{-2}} = 2C_1 + 2C_2\lambda^{-1} \quad (4)$$

where σ^* refers to the reduced stress, λ is the stretch ratio, σ is the stress, $2C_1$ and $2C_2$ are constants which can be determined by the intercepts and slopes of the linear fit in intermediate strain range, for example in the range from 0.5 to 0.7 for our data to avoid strong upturn and strain hardening [39]. This phenomenological

theory is traditionally used to identify the nonideal behaviors between strain and stress. Although the molecular interpretation of $2C_1$ and $2C_2$ terms is still controversial, it is generally accepted that $2C_1$ is associated with strong crosslinks (chemical crosslinks and trapped entanglements), whereas $2C_2$ is related to weak crosslinks (weak entanglements, topological constraints) and is considered an indicator of the departure of the ideal behaviors in polymer network [40,41]. When $2C_2$ is 0, σ^* is independent of λ^{-1} , indicating the case of ideal elastomer (affine network) [42]. The values of $2C_1$ and $2C_2$ of acrylates-cured OS7.8 are listed in Fig. 9b. It can be seen that with the increment of f , $2C_1$ increases monotonously, indicating the increasing of strong crosslinks which is consistent with the measured crosslink density. Besides, the $2C_2$ decreases continuously with the increasing functionality, indicating reduced weak crosslinks content. What's more, with the decreasing of λ^{-1} , namely the increasing of strain, the phenomenon of strain hardening occurs, which is the result of orientation of rubber chains between crosslinks. We can find that crosslinker with higher functionality imparts samples higher degree of strain hardening. At the same time, the onset strain of upturning decreases continuously with increasing f , indicating that high f is beneficial to the orientation of rubber chains, which may be responsible for the high tensile strength and modulus of crosslinked diene rubbers. The occurrence of softening region where stress increases slowly with increasing strain, is an important reason for the low modulus of diene rubber [43,44]. The softening region is thought mainly rooting from the existence of weak crosslinks (namely $2C_2$) [45]. Therefore, reducing $2C_2$ is beneficial to suppressing the softening region and improving tensile modulus. Consequently, higher f imparts elastomer greatly enhanced modulus.

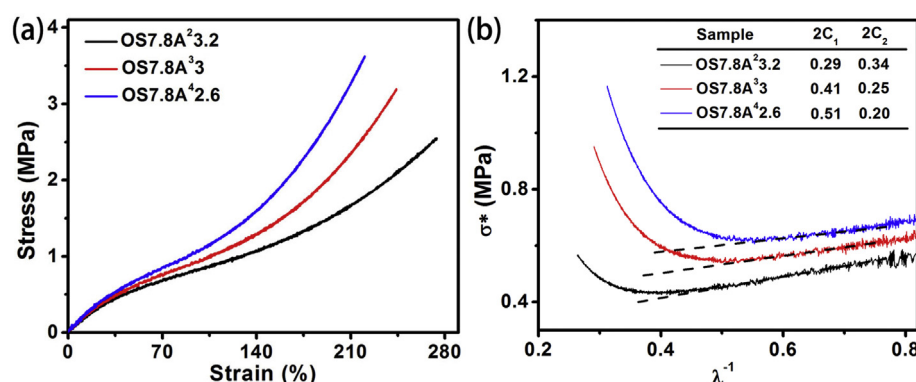


Fig. 9. Typical stress-strain curves (a) and Mooney-Rivlin profiles (b) of OS7.8 cured by acrylates with different functionality.

4. Conclusion

In summary, we demonstrate a new curing chemistry based on oxa-Michael reaction for diene rubbers to achieve the combination of simple curing system and tunable mechanical properties. Specifically, pendent hydroxyls are introduced onto the SSBR via thiol-ene click reaction and the modified SSBR can be effectively cured by various acrylates based on the oxa-Michael reaction without the use of any additional additives. By improving the hydroxyl content, the tensile strength and modulus can be improved while the break strain remains unchanged. Increasing contents and functionality of acrylates can also improve the mechanical properties, especially the striking improvement in modulus. The correlation between mechanical properties and network structure is accordingly achieved, which provides essential base for the future application of the diene rubbers cured by oxa-Michael reaction.

Supporting Information

¹H NMR of OS2.9 and OS7.8, temperature dependence of storage modulus of all samples, tan δ -T curves of samples with tunable functionality and curing parameters of OSxA^wz samples are listed in the Supporting Information. This material is available free of charge via the Internet at <http://pubs.acs.org>.

Notes

The authors declare no competing financial interest.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.polymer.2018.04.039>.

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